

2025 Summer Qual

III.

(a) Using the hint, $A = BC$

$$A^2 = A \Leftrightarrow BCBC = BC$$

$$\Rightarrow \exists D \text{ s.t. } CD = I \quad \because \text{full row rank}$$

$$\Rightarrow BCBCD = BCD$$

$$BCB = B$$

$$\Rightarrow \exists L \text{ s.t. } LB = I \quad \because \text{full column rank}$$

$$\Rightarrow LBCB = LB$$

$$CB = I$$

$$\text{tr}(A) = \text{tr}(A^2) = \text{tr}(BCBC)$$

$$= \text{tr}(CBCB)$$

$$= \text{tr}(I_b)$$

$$= b =: \text{rank}(A)$$



(b) Let G_1 and G_2 s.t. $AG_1A^T = A$,
 $AG_2A^T = A$

$$0 = (AG_1A^T - AG_2A^T)(G_1^T A^T - G_2^T A^T)$$

$$= (AG_1A^T - AG_2A^T)(AG_1A^T - AG_2A^T)^T$$

$$= B^T$$

$$= B^T B$$

$$\Rightarrow B = 0$$

$$\Rightarrow A G_1 A^T - A G_2 A = 0 \leadsto A G_1 A^T = A G_2 A^T$$

So, $A(A^T A)^{-1} A^T$ is invariant to the choice of $(A^T A)^{-1}$.

[2]

$$X_2^* = (I - H_{X_1}) X_2, \quad H_{X_1} = X_1 (X_1^T X_1)^{-1} X_1^T$$

$$Y = X_1 \beta_1 + X_2^* \beta_2 + e.$$

$$(X^T X)^{-1} = \begin{pmatrix} X_1^T X_1 & X_1^T X_2^* \\ X_2^{*T} X_1 & X_2^{*T} X_2^* \end{pmatrix}^{-1}$$

$$= \begin{pmatrix} (X_1^T X_1)^{-1} & 0 \\ 0 & (X_2^{*T} X_2^*)^{-1} \end{pmatrix}$$

$$\hat{\beta}_1 = (X_1^T X_1)^{-1} X_1^T Y$$

$$\begin{aligned} E[\hat{\beta}_1] &= (X_1^T X_1)^{-1} X_1^T [X_1 \beta_1 + X_2 \beta_2] \\ &= \beta_1 + (X_1^T X_1)^{-1} X_1^T X_2 \beta_2 \end{aligned}$$

$$\text{bias of } \hat{\beta}_1 = (X_1^T X_1)^{-1} X_1^T X_2 \beta_2$$

$$\text{var}(\hat{\beta}_1) = \sigma^2 (X_1^T X_1)^{-1}$$

$$\text{var}(\hat{\beta}) = \sigma^2 (X^T X)^{-1}$$

$$= \sigma^2 \begin{pmatrix} \boxed{X_1^T X_1} & X_1^T X_2 \\ X_2^T X_1 & X_2^T X_2 \end{pmatrix}^{-1}$$

$$= \sigma^2 \begin{pmatrix} (X_1^T X_1)^{-1} + (X_1^T X_1)^{-1} X_1^T X_2 (X_2^T (I - H_{X_1}) X_2)^{-1} X_2^T X_1 (X_1^T X_1)^{-1} & \star \\ \star & \star \end{pmatrix}$$

$$\text{var}(\hat{\beta}_1) = \sigma^2 [(X_1^T X_1)^{-1} + (X_1^T X_1)^{-1} X_1^T X_2 (X_2^T (I - H_{X_1}) X_2)^{-1} X_2^T X_1 (X_1^T X_1)^{-1}]$$

$$\text{var}(\hat{\beta}_1) - \text{var}(\hat{\beta}_1)$$

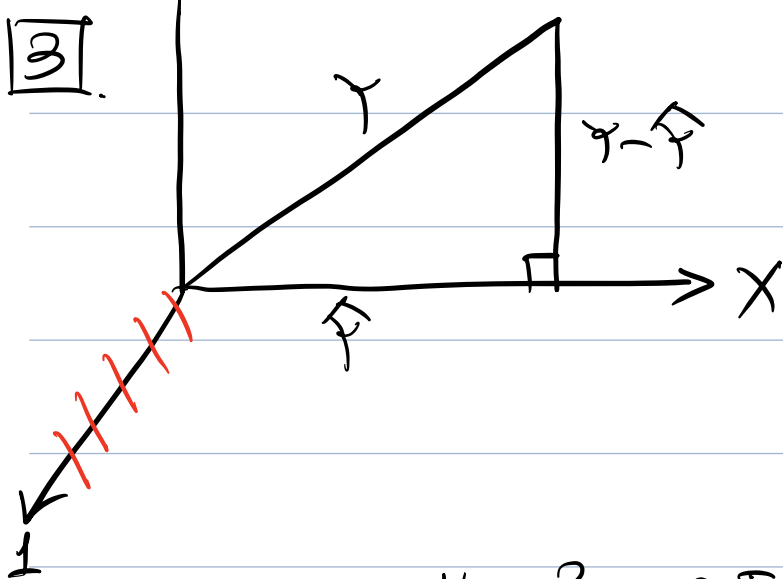
$$= \sigma^2 \underbrace{(X_1^T X_1)^{-1}}_A \underbrace{X_1^T X_2 (X_2^T (I - H_{X_1}) X_2)^{-1} X_2^T X_1}_{B^{-1}} \underbrace{(X_1^T X_1)^{-1}}_{A^T}$$

$I - H_{X_1}$ is symm & idempotent

$$\Rightarrow X_2^T (I - H_{X_1}) X_2 = X_2^T L L^T X_2 \quad \because \text{Cholesky decomposition}$$

$$= (L^T X_2)^T (L^T X_2) \quad \because \text{Positive semidefinite}$$

$$\textcircled{2} \text{var}(\hat{\beta}_1) - \text{var}(\hat{\beta}_1) \succeq 0$$



(a) $SST = \|Y\|^2 = Y^T Y$

$$SSE = \|Y - \hat{Y}\|^2 = Y^T (I - H_X) Y$$

$$SSR = \|\hat{Y}\|^2 = Y^T H_X Y$$

∴ check: $SST = SSE + SSR$

$$Y^T Y = Y^T (I - H_X) Y + Y^T H_X Y$$

□

(b) $H_0: \beta_1 = \dots = \beta_p = 0$ vs. $H_1: \text{not } H_0$

$$F = \frac{SSR/p}{SSE/n-p} = \frac{Y^T H_X Y / p}{Y^T (I - H_X) Y / (n-p)}$$

① $\frac{Y^T H_X Y}{\sigma^2} \sim \chi^2(p, \frac{\beta^T X^T X \beta}{2\sigma^2})$

$$\text{rank}(H_X) = \text{tr}(H_X) = p, \quad \frac{1}{2} \frac{\mu^T H_X \mu}{\sigma} = \frac{\beta^T X^T X \beta}{2\sigma^2}$$

idemp

$$\textcircled{2} \frac{Y^T(I-H)Y}{\sigma^2} \sim \chi^2(n-p)$$

$$\text{rank}(I-H) = \text{tr}(I-H) = n-p,$$

idemp

$$\frac{1}{2} \frac{\partial}{\partial \beta} (I-H) \frac{\partial}{\partial \beta} = \frac{1}{2\sigma^2} \beta^T X^T (X-X) \beta = 0$$

$$\textcircled{3} \text{Independence} \quad HX(I-H) = HX - HX = 0$$

$$F = \frac{\frac{SSR}{\sigma^2} / p}{\frac{SSE}{\sigma^2} / (n-p)} \sim \begin{matrix} H_1 \\ H_0 \end{matrix} F(p, n-p, \frac{\beta^T X^T X \beta}{2\sigma^2})$$

□

□.

$$(a) \max_a \frac{(a^T \hat{\beta})^2}{\sigma^2 a^T (X^T X)^{-1} a} = \frac{\hat{\beta}^T X^T X \hat{\beta}}{\sigma^2}$$

$$\frac{\partial}{\partial a} \frac{(a^T \hat{\beta})^2}{\underbrace{a^T (X^T X)^{-1} a}_{\text{Scalar}}} = \frac{[a^T (X^T X) \hat{\beta}] 2(a^T \hat{\beta}) \hat{\beta} - 2(a^T \hat{\beta})^2 (X^T X)^{-1} a}{[a^T (X^T X)^{-1} a]^2}$$

$$= 0$$

$$\Rightarrow [a^T (X^T X)^{-1} a] \hat{\beta} = (a^T \hat{\beta}) (X^T X)^{-1} a$$

$$\Rightarrow a = \frac{a^T (X^T X)^{-1} a}{a^T \hat{\beta}} X^T X \hat{\beta}$$

$$= c^T X^T X \hat{\beta}$$

$$\begin{aligned} \max_a \frac{(a^T \hat{\beta})^2}{s^2 a^T (X^T X)^{-1} a} &= \frac{c^2 (\hat{\beta}^T X^T X \hat{\beta})^2}{s^2 c^2 \hat{\beta}^T X^T X \hat{\beta}} \\ &= \frac{\hat{\beta}^T X^T X \hat{\beta}}{s^2} \end{aligned}$$

□

(b) Under $H_0: \beta = 0$

$$\frac{\hat{\beta}^T X^T X \hat{\beta}}{ps^2} = \frac{Y^T H_X Y / p}{SSE / (n-p)}$$

$$= \frac{Y^T H_X Y / p}{Y^T (I - H_X) Y / (n-p)} \quad H_X (I - H_X) = 0$$

$$= \frac{\chi^2(p) / p}{\chi^2(n-p) / (n-p)}$$

$$\sim F(p, n-p)$$

□

5.

$$\ell_c(\beta, \sigma^2, \mu_x, \tau^2) = \sum_{i=1}^n \log f(d_i; \mu_x, \tau^2) + \sum_{i=1}^n \log f(y_i | d_i; \beta, \sigma^2)$$

$$\ell_c = -\frac{n}{2} \log \tau^2 - \frac{1}{2\tau^2} \sum_{i=1}^n (d_i - \mu_x)^2 - \frac{n}{2} \log \sigma^2 - \frac{1}{2\sigma^2} \sum_{i=1}^n (y_i - \beta d_i)^2$$

$$\theta^{(k)} = (\beta^{(k)}, \sigma^{2(k)}, \mu_x^{(k)}, \tau^{2(k)})$$

$$y_i | d_i, \theta^{(k)} \sim N(\beta^{(k)} d_i, \sigma^{2(k)})$$

$$\mathbb{E}^{(k)}(y_i) = \beta^{(k)} d_i$$

$$\mathbb{E}^{(k)}(y_i^2) = \sigma^{2(k)} + (\beta^{(k)} d_i)^2$$

$$\mathbb{E}^{(k)}[(y_i - \beta d_i)^2]$$

$$= \mathbb{E}^{(k)}(y_i^2) - 2\beta d_i \mathbb{E}^{(k)}(y_i) + \beta^2 d_i^2$$

$$= \sigma^{2(k)} + (\beta^{(k)} d_i)^2 - 2\beta d_i \beta^{(k)} d_i + \beta^2 d_i^2$$

$$= \sigma^{2(k)} + (\beta^{(k)} d_i - \beta d_i)^2$$

$$= \sigma^{2(k)} + (\beta^{(k)} - \beta)^2 d_i^2$$

$$Q(\theta | \theta^{(k)}) = -\frac{n}{2} \log \tau^2 - \frac{1}{2\tau^2} \sum_{i=1}^n (d_i - \mu_x)^2$$

$$-\frac{n}{2} \log \sigma^2 - \frac{1}{2\sigma^2} \left[\sum_{i=1}^k (y_i - \beta d_i)^2 + \sum_{i=k+1}^n [\sigma^{2(k)} + (\beta^{(k)} - \beta)^2 d_i^2] \right]$$

$$\mu_x^{(k+1)} = \bar{x} = \frac{1}{n} \sum_{i=1}^n d_i \quad \text{by } \frac{\partial Q}{\partial \mu_x}$$

$$\tau^{2(k+1)} = \frac{1}{n} \sum_{i=1}^n (d_i - \bar{x})^2 \quad \text{by } \frac{\partial Q}{\partial \tau^2}$$

$$\sum_{i=1}^k (y_i - \beta x_i)^2 + \sum_{i=k+1}^n (\beta^{(t)} - \beta)^2 x_i^2$$

$$\frac{\partial}{\partial \beta} \Rightarrow -2 \sum_{i=1}^k x_i (y_i - \beta x_i) + 2 \sum_{i=k+1}^n x_i^2 (\beta^{(t)} - \beta) = 0$$

$$\Rightarrow \sum_{i=1}^n \beta x_i^2 = \sum_{i=1}^k x_i y_i + \beta^{(t)} \sum_{i=k+1}^n x_i^2$$

$$\Rightarrow \beta^{(t+1)} = \frac{\sum_{i=1}^k x_i y_i + \beta^{(t)} \sum_{i=k+1}^n x_i^2}{\sum_{i=1}^n x_i^2}$$

$$\frac{\partial}{\partial \sigma^2} = -\frac{n}{2\sigma^2} + \frac{1}{2\sigma^4} \left[\sum_{i=1}^k (y_i - \beta x_i)^2 + \sum_{i=k+1}^n \sigma^{2(t)} + (\beta^{(t)} - \beta)^2 x_i^2 \right]$$

$$\sigma^{2(t+1)} = \frac{1}{n} \left[\sum_{i=1}^k (y_i - \beta^{(t+1)} x_i)^2 + \sum_{i=k+1}^n \sigma^{2(t)} + (\beta^{(t)} - \beta^{(t+1)})^2 x_i^2 \right]$$

□